When Self-Consistency Backfires: Majority Vote Hurts the Majority of Hard Science Problems for Small LLMs

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Abstract

Self-consistency (SC) via majority vote reduces per-problem accuracy on most GPQA Diamond problems for small instruction-tuned models: 56.6% of problems for Qwen2.5-7B and 65.7% for Llama-3-8B. The obvious remedy is a verifier-free confidence gate that decides which problems to vote on. This version reports that the most natural repair to that remedy also fails, and separates three signal failures that v1 treated as one. A token-entropy gate fails for a measurement reason rather than a substantive one: a per-token average over a chain averaging 602 tokens is dominated by fluency, and measured, the chain average sits 0.0005 nats from the average over non-answer tokens and 0.1695 nats from the answer token's own value. We therefore measure confidence at the answer span itself, where that dilution cannot apply. On Qwen2.5-7B-Instruct-Turbo, 198 problems at 64 samples each, a sample whose answer contradicts its own problem's plurality still emits that answer at a median margin of 20.52 nats, with 75.7% of such samples above 10 nats. Both quantities were pre-registered with thresholds fixed in advance and tested once on 69 problems that no exploratory analysis had read; both passed. The unit is the whole result: pooled across the benchmark the same margin does separate correct from incorrect samples, by $+0.0604$ on the fraction above 10 nats $[+0.0183, +0.1017]$, excluding zero; measured within each problem and paired across its own samples it does not, at -0.0168 $[-0.0527, +0.0182]$, crossing zero. So the claim is not that token log-probabilities carry no information, but that a signal with genuine across-question discrimination is close to useless for the within-question decision self-consistency actually poses. The plurality-agreement gate's failure, unlike the other two, remains without a mechanism, and we report it as an open problem. These new claims rest on one model: a registered second-model replication was sampled and could not be evaluated, and we report that rejection rather than the result. We separately report that on hosted serverless inference at a small budget, three reasoning-native models could not be evaluated at all, for three distinct and separately measured reasons; all three are downloadable, so this bounds what a metered per-token API buys rather than what is knowable.

Changes from v1

Peer-review status. The COLM 2026 Workshop on Efficient Reasoning accepted v1 of this paper (arXiv:2608.11403), whose content is the original Introduction, Setup, Results and Discussion. **No reviewer saw the material added in v2**: the answer-token margin and the commitment result, the reasoning wall, the rejected second-model replication, the reconstructed estimator, and the corrections section. v2 also *corrects* the accepted version: Section 4.4 replaces the accepted paper's mechanism section, and one Discussion sentence is withdrawn as stated. Readers relying on peer review should treat the accepted claims and the new ones differently, and the list below marks which is which.

For readers who cited the original. **The headline results are unchanged: no v1 result was recomputed, re-estimated, or withdrawn.** What moved is the explanation of why the gates fail.

- **Corrected.** Section 4.4's v1 thesis, "confidence does not track correctness", inferred one mechanism from two gates failing together. The failures have different causes and one of them is a measurement artifact.
- **Corrected, narrowed to a unit.** v1's Discussion claim that "confidence on these problems does not indicate correctness" is false as stated across questions: pooled, the answer-token margin separates correct from incorrect by $+0.0604$ $[+0.0183, +0.1017]$. Within a problem, paired, it does not: -0.0168 $[-0.0527, +0.0182]$. The defensible claim is the within-question one.
- **Partly answered, and rescoped.** v1's "we do not test reasoning-native models, which we flag as the central open question" is no longer open in the same way: on hosted serverless inference at a small budget it is unmeasurable (Section 7). It remains open for the open-weights route.
- **Separated.** The entropy gate's failure was grouped with agreement's under one known-mechanism heading. It was reading a diluted statistic, which is miscounted evidence rather than evidence about confidence.
- **Added.** The answer-token margin (the "richer signal" v1's Limitations named as future work), the reasoning wall, a registered replication that this paper's own comparability rule rejected, a reconstructed few-sample estimator, and a corrections section.

v2 closes a limitation v1 stated about itself. v1 wrote that evaluating final-answer margins "requires per-token log-probability arrays, which our pipeline did not store". This version stores them and evaluates that signal. It does not behave differently in the way that sentence hoped.

Data availability

Derived, question-free data reproducing every number in this paper is archived open access at 10.5281/zenodo.21933418 (CC BY 4.0). The raw per-sample store, including model chains of thought, is archived at 10.5281/zenodo.21933422 as a **restricted, request-access** record on the same terms as GPQA (Rein et al., 2024); it is not open data, for the reason given in Section 7.1.

1 Introduction

Inference-time compute is a primary lever for LLM reasoning, and self-consistency—sample several chains of thought, return the plurality answer—is among the simplest and most widely used techniques in this family (Wang et al., 2023). It is commonly assumed to be a low-risk accuracy boost: sample more, vote, do at least as well.

That assumption fails on hard problems. When a model places its highest probability on an incorrect answer across independent samples, more samples only entrench the wrong vote. We call this *backfire*: $\text{mv\_gain} = \text{MV\_acc}(64) - \text{MV\_acc}(1) < 0$, where $\text{MV\_acc}(N)$ is the expected accuracy of a majority vote over N samples. We then ask the question a cost-conscious practitioner would ask: can a cheap, verifier-free signal computed from a few samples tell you which problems to vote on and which to skip? We test the two most natural signals, plurality agreement and token-level entropy, and find both fail, and we identify why.

That self-consistency can hurt is not itself new, and the underlying miscalibration is well documented (Guo et al., 2017; Kadavath et al., 2022). Our contribution is to quantify, pre-register, and confirm how often it happens on a full hard benchmark, across two model families, and to show that two cheap gates cannot prevent it.

Contributions:

- On all 198 GPQA Diamond problems, majority voting backfires on the majority of problems for two model families (56.6% and 65.7%), with Qwen the primary demonstration and Llama corroborating from a near-chance baseline. The effect was pre-registered on a 151-problem held-out split and confirmed (all four hypotheses passed).
- We show the per-problem routing headroom is real: a grid oracle (best N per problem across $\{1, 2, 4, 8, 16, 32, 64\}$) marks a 14 to 17 point ceiling above $N = 1$. Neither a plurality-agreement gate nor a token-entropy gate reaches it, and neither moves accuracy more than 0.002 from fixed-budget voting at $N = 64$.
- We give the mechanism: agreement does not track correctness, and for the weaker model higher agreement does not indicate higher accuracy.

New in v2. v1 tested two such signals, plurality agreement and token-level entropy, found both fail, and offered one explanation for both. This version tests a third, the answer-token log-probability margin, which v1's own Limitations named as the obvious next candidate and could not evaluate because its pipeline stored only a mean entropy scalar. We collect the per-token arrays, measure the margin at the answer span, and report that it also fails. The three failures do not share an explanation, and separating them is the main interpretive change in this version: one is a measurement artifact, one has a mechanism we can state and test, and one remains unexplained.

2 Setup

Dataset and design. We use the full GPQA Diamond benchmark (Rein et al., 2024), 198 graduate-level multiple-choice questions in biology, chemistry, and physics. We adopt a pre-registered confirmatory design: 47 problems are **exploratory** (the hypotheses below were generated from them), and the remaining 151 are a **confirmatory** test set. The hypotheses and thresholds were locked and git-tagged before any confirmatory analysis, at tag `backfire-prereg-v1.0` in the repository at `https://github.com/u7k4rs6/self-consistency-backfire`. We report exploratory, confirmatory, and pooled (198) values throughout, but decide PASS/FAIL on the confirmatory set only.

Models. Two instruction-tuned models from different families, served on Together AI: Qwen2.5-7B-Instruct-Turbo and Meta-Llama-3-8B-Instruct-Lite. Both are small (7 to 8B) and non-reasoning. $N = 64$ samples per problem, temperature 0.7, single locked prompt template, byte-identical across all runs as verified by SHA-256. Five-pass answer extraction; parse rate 99.5% (Qwen) and 98.6% (Llama). Llama has exactly 64 stored samples per problem. For Qwen the 151 confirmatory problems have exactly 64 at this temperature and the 47 exploratory problems have 65 to 72, retained from earlier sampling runs, so pool size and split membership coincide exactly; three further problems hold side-test samples at other temperatures, which the analysis filters out. $\text{MV\_acc}(N)$ draws subsets of size $\min(N, n)$ without replacement from all n stored samples for a problem, so for those 47 the estimate at $N = 64$ is taken over a larger pool than for the rest. Reclassifying every problem from its first 64 samples leaves the backfire count unchanged at 112 of 198, with no problem crossing the backfire boundary (`scripts/verify_truncation_invariance.py`).

Metrics. $\text{MV\_acc}(N)$ is expected majority-vote accuracy over N samples, ties broken uniformly at random independent of ground truth,[1] estimated by Monte Carlo over subsets. Backfire is $\text{mv\_gain} < 0$.

Routing and gates. The **oracle** routes each problem to the N that maximizes majority-vote accuracy across $\{1, 2, 4, 8, 16, 32, 64\}$ using ground truth; it is a theoretical upper bound, not a deployable method. The **agreement gate** returns the probe plurality when its fraction over

[1] This v1 sentence is retained verbatim. The implementation breaks ties lexicographically rather than at random, and the published numbers follow the implementation; see disclosure 9 in Section 10, which quantifies the difference as a third-decimal effect that changes no conclusion.

k samples is at least τ, else votes at $N = 64$. The **entropy gate** returns the probe plurality when the mean per-token entropy over $k = 4$ probe samples falls below a threshold, else votes at $N = 64$. Both gates are verifier-free; ground truth is used only to score the result. Logprobs are available for all 151 confirmatory problems but not for the 47 older exploratory ones, so the entropy gate is evaluated on the confirmatory set. Its threshold is selected on that same set by an argmax over 40 candidate values, maximizing capture against the binary $\{N = 1, N = 64\}$ oracle relative to an MV_acc(64) baseline (Qwen: 0.113, Llama: 0.139).

Uncertainty. 95% confidence intervals are problem-level bootstrap (1000 iterations, seed 42).

2.1 New in v2: the answer-token margin

v1 retained only a mean per-token entropy scalar, which is why it could not evaluate a margin. This version re-samples Qwen2.5-7B-Instruct-Turbo on all 198 problems at $M = 64$, cap 2048, requesting log-probabilities at depth 5, and stores the full response object: 12,672 samples.

The **answer-token margin** is the log-probability of the emitted option letter minus that of the highest-scoring alternative option letter, at the answer position. The provider returns the top k alternatives per token, $k = 5$ in practice, and five slots need not contain every option letter. With two or more option letters present the margin is *measured*: the second-highest present option is a returned value and any absent letter lies at or below the smallest returned value, so no absent letter can outrank it. With fewer than two present it is *right-censored* at the top letter minus the smallest returned log-probability, and recorded as a bound. It is never imputed: filling a missing letter at the censoring bound would understate the margin exactly on the problems where the model is most certain, which are the ones a confidence gate cares about most. Of 12,672 samples, 12,344 margins are measured, 265 right-censored, and 63 have no margin at all because no answer was extracted; those 63 are the same samples behind the 0.9950 answer rate and are never scored as incorrect.

The new run's answer rate is 0.9950 $[0.9936, 0.9961]$ against v1's 0.9946 $[0.9932, 0.9958]$ for the same model, so the two runs describe the same output population, not merely the same prompts. Exposure was tracked per analysis: 129 problems were read by some exploratory analysis and 69 were not. The claims in Section 4.5 were registered with thresholds fixed before those 69 were examined, and were examined once.

3 Pre-Registered Confirmatory Results

All four pre-registered hypotheses pass on the 151 confirmatory problems (Table 1). Thresholds were fixed before any confirmatory analysis and set with margin on the permissive side of the exploratory point estimates rather than at them, so each is a genuine prediction rather than a restatement. PH4 had no exploratory estimate, since logprobs were not collected for the exploratory split, and was given the same 10% ceiling as PH2. PH2 and PH4 were pre-registered against a binary $\{N = 1, N = 64\}$ oracle; the grid oracle results in Sections 4.2 and 4.3 use a more generous ceiling and are reported separately as post-hoc analysis.

On PH2, added in v2 and leaving the published verdict untouched. PH2 is recorded above as PASS, which is the correct application of its pre-registered decision rule: that rule is stated on the point estimate, and the point estimates are 0.8% and -1.6%, both below the 10% ceiling. The intervals around them are wide enough to be consistent with capture well above that ceiling. The verdict as published stands and is not revised here; what v2 adds is that the interval, not the point estimate, is the honest measure of what this hypothesis established, and it established less than the PASS alone suggests.

Hyp	Prediction (both models)	Qwen2.5-7B	Llama-3-8B	Result
PH1	backfire rate $\geq$ 33%	60.3% [53.0, 68.2]	65.6% [58.3, 73.5]	PASS
PH2	agree gate capture $\leq$ 10%	0.8% [−89.1, 68.1]	−1.6% [−92.9, 74.2]	PASS
PH3	top-agree-bin acc. $\leq$ 70%	51.2% (n=43)†	14.3% (n=21)†	PASS
PH4	entropy gate capture $\leq$ 10%	0.5%†	0.9%†	PASS

Table 1: Confirmatory hypotheses ($n = 151$). Thresholds locked and git-tagged before analysis. Captures in this table are measured against the binary $\{N = 1, N = 64\}$ oracle relative to an MV_acc(64) baseline, which is why they are much smaller than the grid-oracle captures reported in Section 4.3. Bracketed values are 95% bootstrap intervals (problem-level, 1000 iterations, seed 42); † marks quantities for which no interval was computed. PH2's intervals are wide enough to be consistent with capture well above the 10% threshold: the pre-registered decision rule is on the point estimate, and we report the intervals alongside it.

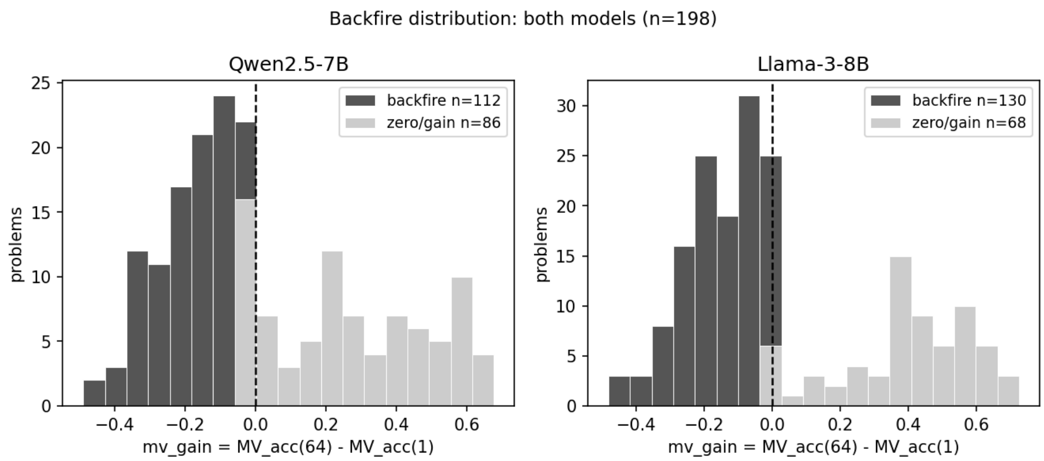

Figure 1: mv_gain $=$ MV_acc$(64) -$ MV_acc(1) per problem. Dark bars are backfire (mv_gain < 0).

4 Results (Pooled, 198 Problems)

4.1 Backfire affects the majority of problems, precisely estimated

Majority voting reduces per-problem accuracy on most problems for both models (Figure 1): the pooled backfire rate is 56.6% (95% CI [49.5, 63.6]) for Qwen and 65.7% ([59.1, 71.7]) for Llama. Expanding from 47 to 198 problems roughly halved the interval width, and both rates sit well above the 33% threshold.

The aggregate and per-problem pictures diverge. Voting barely moves aggregate accuracy (Qwen 0.342 to 0.369, Llama 0.273 to 0.313) while harming the majority of problems individually. An average that looks flat or mildly positive can hide widespread per-problem harm. The worst single problem loses 47 points (Qwen) or 46 (Llama); a few gain as much as 66 or 70. The asymmetry is real but rare: large gains exist, yet most problems backfire.

Backfire is not uniform across domains (Table 2). Chemistry is the dominant contributor: it accounts for 93 of the 198 problems and shows the highest backfire rate for both models (66.7% Qwen, 68.8% Llama). Physics is intermediate (50.0% Qwen, 67.4% Llama), and biology is least affected, though its small problem count ($n = 19$) makes that estimate noisy. We report this breakdown descriptively: the per-domain counts, biology especially, are too small to support claims about why backfire concentrates where it does.

Domain	Qwen n	Qwen backfire	Llama n	Llama backfire
Biology	19	36.8%	19	42.1%
Chemistry	93	66.7%	93	68.8%
Physics	86	50.0%	86	67.4%
Pooled	198	56.6%	198	65.7%

Table 2: Per-domain backfire rate (pooled 198). Chemistry shows the highest backfire rate in both models and accounts for nearly half of all problems.

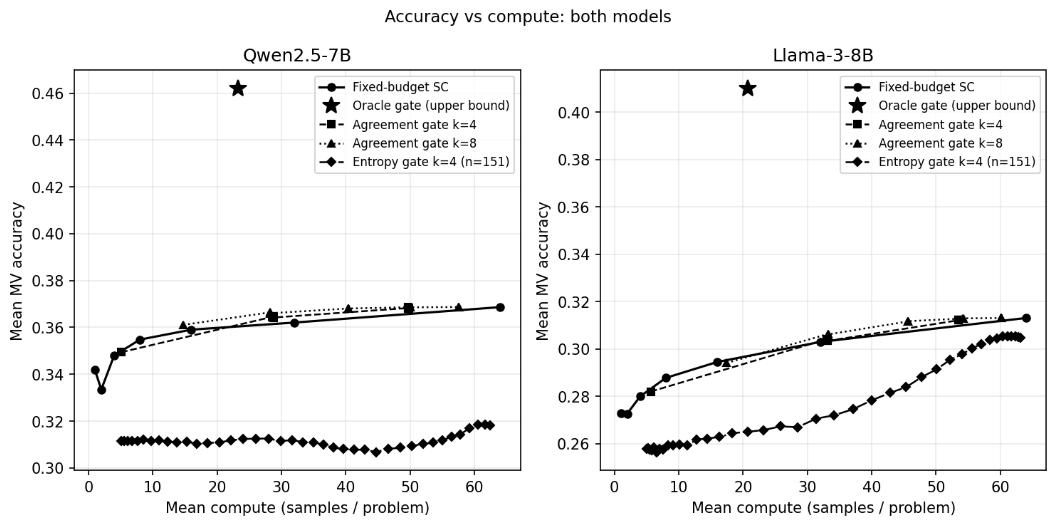

Figure 2: Accuracy vs mean compute: fixed-budget voting, the oracle upper bound, and the verifier-free agreement and entropy gates. The plotted oracle is the binary $\{N = 1, N = 64\}$ oracle (0.462 Qwen, 0.410 Llama), not the more generous grid oracle whose higher values are quoted in Section 4.2. The entropy series is computed on the confirmatory 151 only, a harder subset than the pooled 198 used for the other series, so its lower position does not by itself indicate a worse method.

4.2 The oracle upper bound is real but not reachable

A grid oracle that routes each problem to the best majority-vote accuracy across all N in $\{1, 2, 4, 8, 16, 32, 64\}$ reaches 0.482 (Qwen) and 0.439 (Llama), a ceiling 14 points above $N = 1$ for Qwen and 17 points above for Llama. This is a more generous bound than a binary $\{N = 1, N = 64\}$ oracle, because it allows any compute level per problem. It marks how much accuracy a perfect per-problem routing decision could recover. It is an upper bound, not a method (Figure 2). The question this raises is how much of that ceiling is reachable without ground truth, which Section 4.3 answers by testing two verifier-free gates.

4.3 Two verifier-free gates fail to capture it

Neither cheap gate recovers the headroom. The agreement gate ($k = 8$, $\tau = 0.75$) reaches 0.368 accuracy for Qwen and 0.312 for Llama, against 0.369 and 0.313 for fixed-budget voting at $N = 64$, differences of 0.0006 and 0.0014. We report these magnitudes rather than a significance claim, having computed no paired test. Against that ceiling, the agreement gate captures 18.7% of available headroom for Qwen and 23.4% for Llama (headroom is measured against the grid oracle of Section 4.2, not the binary oracle in Table 1): non-trivial fractions of a generous ceiling, yet accuracy remains flat. The gate runs at a mean of 40.4 samples per problem for Qwen and 45.6 for Llama, which places it between two fixed-budget grid points. Against $N = 64$ it reaches 0.3680 and 0.3117, within 0.0006 and 0.0014 of that baseline, on 63% and 71% of the samples. Against $N = 32$, which scores 0.3621 and 0.3030, it is higher by 0.0060 and 0.0087 but spends 26% and 43% more samples. What the gate buys relative to a flat $N = 64$ is therefore compute rather than accuracy. Those capture figures are measured against MV_acc(1), so they credit the gate with headroom that voting alone already recovers. Measured instead against MV_acc(64), which isolates what the routing decision adds over simply voting, the capture is PH2 in Table 1: 0.8% for Qwen and -1.6% for Llama. Agreement routing therefore contributes essentially nothing to accuracy that fixed-budget voting does not already provide. The gate spends k probe samples on every problem, then either returns the probe plurality (if the plurality fraction $\geq \tau$) or proceeds to $N = 64$. Every gate-versus-baseline comparison we report uses fixed-budget voting at a flat $N = 64$; we do not compute a compute-matched baseline.

The entropy gate is evaluated entirely within the confirmatory 151, since logprobs exist only for that split. Measured consistently within it (the threshold was also selected on this set, so these are in-sample figures), the gate captures 0.2% of the grid-oracle headroom for Qwen and 31.2% for Llama, with gate accuracy of 0.318 (Qwen, threshold 0.113) and 0.306 (Llama, threshold 0.139). Because this capture is computed on a different problem set from the agreement-gate figures above, the two should not be read as like-for-like: the entropy percentages are confirmatory-151 values and the agreement percentages are pooled-198 values.

The like-for-like baselines are therefore the confirmatory-151 fixed-budget accuracies, and the two models differ. For Qwen these are MV_acc$(1) = 0.3182$ and MV_acc$(64) = 0.3179$, against a gate accuracy of 0.3184: the gate matches single-sample accuracy to three decimals, so it does nothing rather than actively hurting, and the apparent five-point drop against the 0.369 quoted above is entirely the pooled-versus-confirmatory mismatch. For Llama they are 0.2518 and 0.3046, against a gate accuracy of 0.3055: there the gate is level with full voting and well above single-sample. One further point: Qwen's confirmatory MV_acc(64) sits marginally below its MV_acc(1), so on this split voting at $N = 64$ is very slightly worse than not voting at all. The gap is small and we do not lean on it; it complements the pooled aggregate reported above (0.342 to 0.369) by showing the per-problem backfire effect surfacing faintly in the aggregate.

Mean per-token entropy modestly predicts which problems backfire (AUC 0.631 for Qwen, 0.523 for Llama), but predictive signal does not translate into routing accuracy because the gate cannot act on that signal without misclassifying enough problems to erase the gain. Across all 22 operating points swept ($k \in \{4, 8\}$ and τ from 0.50 to 1.00 in steps of 0.05), none beats fixed-budget meaningfully (Figure 3). The obvious agreement signal and an uncertainty signal both fail.

4.4 Why: three signals, three different failures

This subsection replaces v1's "Why: confidence does not track correctness". v1 offered a single account for two failing gates. With a third signal measured, that account no longer holds as stated: the three failures have three different statuses, and only one of them is explained by miscalibration.

Signal 1: plurality agreement. Fails, and we cannot say why. The evidence is unchanged (Table 3, Figure 4). Even the highest-agreement bin is far from reliable: Qwen's plurality is correct 52.5% of the time there, and Llama's only 28.6%, lower than its own low-agreement bin, and Llama's accuracy is not monotone in agreement. What v2 changes is the status of the explanation, not the evidence. Plurality agreement is computed over *answers*, not over tokens: it never touches a log-probability, so neither dilution nor answer-token saturation can apply to it, and the account developed below for the other two signals says nothing about this one. **Agreement's failure therefore remains without a mechanism, and we state that as an open problem rather than absorbing it.** Calling it an instance of general overconfidence (Guo et al., 2017; Kadavath et al., 2022) is a label rather than an account: it does not predict that Llama's top bin should be *worse* than its bottom one, and it does not say what would have to be true for agreement to work on some other benchmark. This

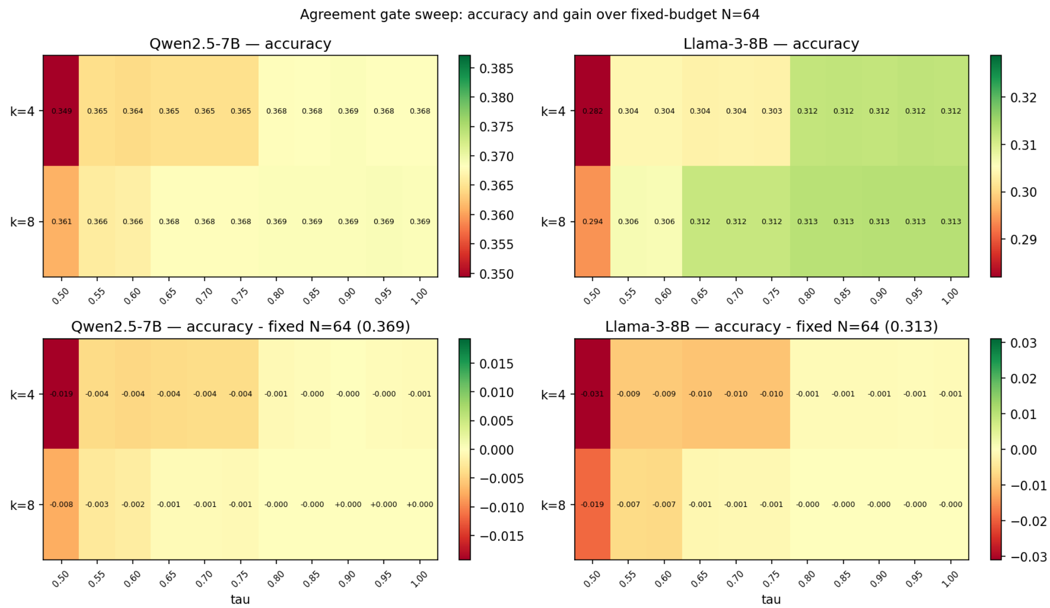

Figure 3: Agreement-gate accuracy across the (k, τ) sweep. The lower row plots the difference from fixed-budget voting at a flat $N = 64$. No operating point wins.

is the largest unresolved question in the paper, and it is as unresolved in v2 as in v1. The difference is that v1's framing made it look answered.

Signal 2: mean token entropy. Miscounted evidence, not evidence about confidence. The entropy gate averaged a per-token quantity over the whole chain. In the margin-v1 store that chain averages 602.21 tokens, of which the answer token is *one*. Measured over 2,560 samples on 40 problems drawn with a recorded seed, using per-token entropy over the returned top-5 alternatives, the mean over the whole chain is 0.2232 nats, the mean over non-answer tokens is 0.2237, and the value at the answer token is 0.0536 with a median of exactly 0.0000. The chain average therefore sits 0.0005 nats from the non-answer average and 0.1695 nats from the answer token's own value: the gate's statistic is, to four decimal places, a measurement of the prose. A gate thresholding it is thresholding fluency. **This is a measurement artifact and should not be reported as a fact about confidence.** That averaged confidence is diluted by high-confidence filler is established rather than new: it is the motivating premise of relevance-weighted uncertainty estimation (Duan et al., 2024) and the target of recent length-invariant estimators (Vashurin et al., 2025). The related pathology in machine translation is that sentence-level model probability produces both a beam problem and a brevity problem, which Murray & Chiang (2018) trace to label bias and correct with a sentence-level term, finding a per-word reward slightly better than length normalization; that is a different mechanism from ours and we cite it as a precedent for correcting a length-coupled score, not as the same finding. Measuring at the answer span instead is likewise established. **We adopt the fix rather than proposing it**, and its role here is narrow: it closes off "you measured in the wrong place" as an explanation for the next result.

Signal 3: the answer-token margin. Fails on saturation, with a mechanism. Measured at the answer token, where dilution cannot apply, the signal is saturated. Section 4.5 gives the registered test. What it supports, stated as an inference rather than an observation: *the answer token behaves as a near-deterministic readout of a chain that has already committed*. We observe saturation downstream, at the answer position, and infer commitment upstream; we do not observe the commitment itself. The saturation is *consistent with* a chain that fixed its answer before the answer token was emitted, and it *supports* that reading over one in which the answer token is where the model decides, but it does not establish when or where

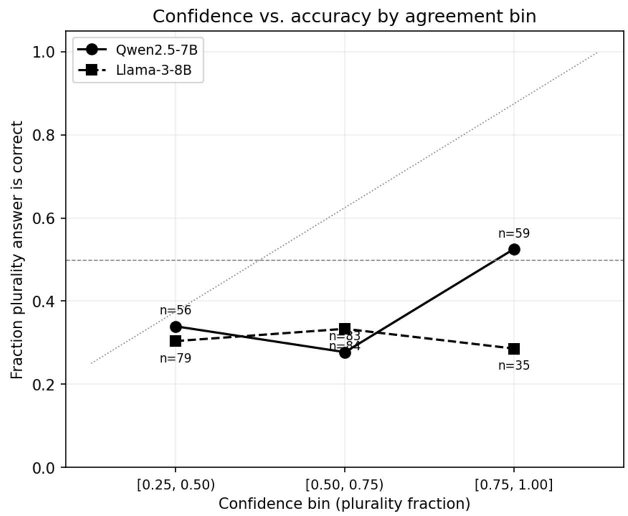

Figure 4: Fraction of plurality answers correct by agreement bin. Both models fall far below the calibration diagonal; Llama's highest-agreement bin is less accurate than its lowest, with no monotone trend across bins.

Confidence bin	Qwen n	Qwen frac correct	Llama n	Llama frac correct
[0.25, 0.50)	56	33.9%	79	30.4%
[0.50, 0.75)	83	27.7%	84	33.3%
[0.75, 1.00]	59	52.5%	35	28.6%

Table 3: Calibration by agreement bin (pooled 198). Confidence is the plurality fraction over all samples.

the decision was made. A sample whose answer contradicts the plurality of its own problem still emits that answer at a median 20.52 nats, odds of roughly 800 million to one against the nearest alternative option. Dissenting samples are as committed as agreeing ones. On this reading the variance self-consistency exploits is not expressed at the answer token, and is more plausibly carried by which chain got written, though that locus is inferred here and not measured. This is a mechanism rather than a restatement because it is falsifiable and was registered in advance: had dissenting samples been measurably less certain than agreeing ones, the thresholds in Section 4.5 would have failed.

What this does to v1's thesis. One of the three failures is not about confidence, one is about confidence in a way v1 did not state, and one is unexplained. **v1's "confidence does not track correctness" is retired as a unifying thesis**, and this subsection must not be read as though every failure now has an account. Two of three do.

4.5 The commitment result, registered and held out

The margin claims were registered with thresholds fixed, then tested once on the 69 problems no exploratory analysis had read.

The holdout reproduces the exposed set closely, 20.52 against 20.62 and 0.757 against 0.749; only the holdout figures were registered.

The two units, measured. The margin is not devoid of information about correctness, and the difference between the two units is not an argument we make about the result: it is the result. Pooled at sample level over all 198 problems (4,283 correct and 8,322 incorrect samples with measured margins), the fraction above 10 nats is 0.8228 for correct samples

Id	Quantity, over dissenting samples	Estimate	95% lower	Threshold
MD1	per-problem median margin (nats)	20.5232	18.8376	15.0
MD2	per-problem fraction above 10 nats	0.7567	0.7105	0.60

Table 4: Pre-registered descriptive claims on the 69 unexposed problems. Both PASS. One-sided 95% lower bounds, cluster bootstrap over problems. 64 of 69 problems carried at least three dissenting samples; 5 were excluded by the registered rule and are counted, not imputed. Answer rate on the store is 0.9950 $[0.9936, 0.9961]$.

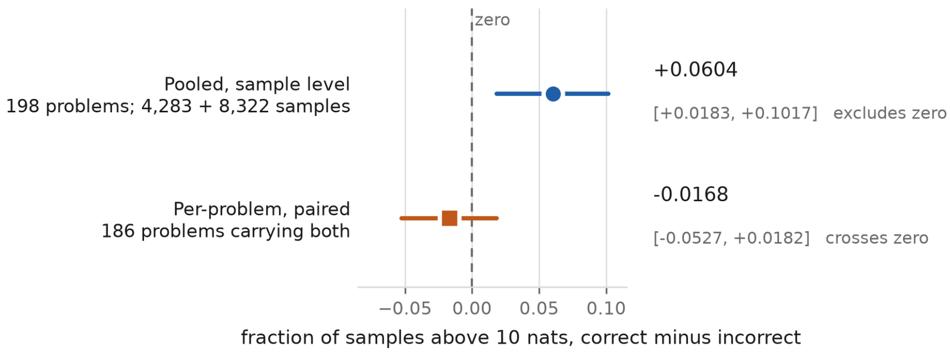

Figure 5: The same margin, the same statistic, two units. **Pooled at sample level** across all 198 problems (4,283 correct and 8,322 incorrect samples carrying a measured margin), correct samples exceed 10 nats more often than incorrect ones and the interval excludes zero: this answers “does the margin carry information about correctness across the benchmark?”, and the answer is yes. **Per-problem and paired**, over the 186 problems carrying at least one sample of each kind, the difference cannot be distinguished from zero: this answers “within a single problem, does the margin say which of several disagreeing samples to trust?”, and the answer is no. The 12 problems absent from the paired row are all-correct (2) or all-incorrect (10), where a paired difference is undefined. **The two are not in conflict**: they are different questions, and only the second is the one a self-consistency router has to answer. Intervals are 95% cluster bootstrap over problems.

against 0.7624 for incorrect, a difference of $+0.0604$ $[+0.0183, +0.1017]$, which excludes zero. Per-problem and paired, over the 186 problems carrying at least one of each, the mean difference is -0.0168 $[-0.0527, +0.0182]$, which crosses zero (Figure 5). The population differs because a paired difference is undefined where a problem has no correct or no incorrect sample: 2 problems are all-correct and 10 all-incorrect, and those 12 contribute to the pooled figure and not the paired one.

We therefore do not claim that token log-probabilities are uninformative, and any reading of this paper reaching that conclusion has overshot. Across questions the margin carries a real, small signal; within a question it does not carry one we can detect, and within a question is where a router stands. This makes the reconciliation with recent work a measurement rather than an argument. Kumaran (2026) reports that calibrated log-probability confidence behaves as an answer-evidence signal coupled to correctness, at AUROC 0.62 to 0.80. That quantity, a temperature-scaled softmax over the option letters, is ours at the same locus. Its unit is not: every result there is trial-level across questions, with one answer drawn per question, and the design never conditions on samples that disagree with each other because it never has two samples of one question to compare. Our pooled figure reproduces the direction that paper reports; our paired figure is the one it never measured.

No margin gate was built, and this is not an evaluation of one. We measured the signal’s distribution and report that it does not separate the samples a within-question router would have to separate. **That is evidence against the signal supporting the required decision, not**

an empirical evaluation of a deployed margin gate, and Section 4.3 accordingly remains a two-gate result.

The objection this invites should be stated rather than left for a reader to find: **a signal with weak paired discrimination can still support useful routing if the routing rule is nonlinear** in it. A threshold that fires only in the extreme tail, a rule combining the margin with agreement or with length, or a per-problem rather than per-sample criterion could each extract value that a paired mean difference does not see. None of those is tested here. What we show is that the simplest reading, that a more certain sample is more likely to be the right one within its own problem, does not hold; what we do not show is that no rule built on this signal can work.

5 Discussion

Why backfire occurs. On hard problems the model's sampling distribution concentrates on a wrong answer, so voting locks in the error. Backfire is not a sampling artifact; it reflects the per-problem answer distribution. GPQA distractors are designed to be plausible (Rein et al., 2024), which amplifies the effect.

Why the gates fail (revised in v2). v1 wrote: "Both gates read confidence (agreement, or low entropy), but confidence on these problems does not indicate correctness." That sentence is withdrawn as stated. The entropy gate was not reading confidence; it was reading a diluted average dominated by prose. The margin, which does read confidence at the right place, fails because the answer token is saturated whether or not the sample agrees with its own plurality. And the agreement gate's failure is explained by neither account. Three signals, three statuses, set out in Section 4.4.

Positioning. Prior work established that self-consistency helps on benchmarks where models have higher baseline accuracy (Wang et al., 2023); we show it hurts the majority of problems on a hard one. Repeated-sampling analyses study coverage under an oracle metric (Brown et al., 2024); we instead decompose the gap between realizable majority vote and the oracle. Adaptive-consistency early stopping (Aggarwal et al., 2023) is essentially our agreement gate, validated on easier datasets where backfire is rare; our negative result shows agreement stability is insufficient on hard problems. Closest to our work, Chen et al. (2024) show that majority-vote accuracy can rise and then fall as the number of LM calls grows, attribute this non-monotonicity to a mixture of easy and hard queries within a task (more calls help the easy ones and hurt the hard ones), and use that structure to estimate, from a small number of samples, the call count that maximizes aggregate performance. We differ in two ways. We measure the fraction of individual problems harmed rather than the shape of the aggregate curve, which matters because aggregate accuracy here looks nearly flat (0.342 to 0.369 for Qwen) while a majority of problems degrade underneath it. And our gate results sit in tension with few-sample estimation of an optimal call count. v1 put this as a claim about a uniformly hard benchmark; the stored data does not support that description for these models (disclosure 10), so the claim is narrower than v1 stated. On this benchmark, whose per-problem difficulty is measurably mixed rather than uniform, the verifier-free signals such estimation would rely on do not recover the headroom. Tan et al. (2025) study self-consistent errors, where a model repeats the same wrong answer across samples, and find that consistency-based detectors, of which semantic cross-checking is one example (Zhang et al., 2023), fail on exactly those cases, which leads them to argue for an external verifier; their setting is error detection rather than compute allocation, but the conclusion converges with ours. The external-signal direction this motivates, trained verifiers and process reward models (Cobbe et al., 2021; Lightman et al., 2024), is the natural way to reach the oracle headroom, which our two verifier-free gates cannot.

Implications. On hard inputs, do not assume self-consistency is safe, and do not expect agreement or token entropy to tell you when it is; we tested both and both fail. Recovering the headroom likely requires a signal external to the model's own samples.

6 Limitations

Llama is near chance. Llama-3-8B's single-sample accuracy on full GPQA Diamond is 0.273, just above the 0.25 random baseline, so its backfire is less surprising. Qwen (0.342, clearly above chance) is the stronger demonstration; Llama corroborates the direction.

One benchmark. All results are on GPQA Diamond, graduate-level science. Other domains and easier difficulty regimes may differ.

Two small, non-reasoning models. Reasoning-native models require different infrastructure. We attempted a preliminary evaluation of a reasoning-native model with hidden chain-of-thought and found that a substantial fraction of samples exhausted the output token budget on hidden reasoning before producing a visible answer. This truncation made majority-vote accuracy incomparable to our non-reasoning models. A proper evaluation of reasoning-native architectures therefore requires larger inference budgets and explicit handling of hidden CoT output length, and is left to future work. Whether backfire shrinks or persists for such models remains the central open question.

Subset representativeness. The original 47-problem subset was easier than the full benchmark (Qwen $N = 1$ accuracy 0.418 vs 0.342 pooled), which is why we report on all 198.

Monte Carlo boundary sensitivity. $\mathrm{MV\_acc}(N)$ is estimated by Monte Carlo, so a problem whose mv_gain is exactly zero can fall on either side of the backfire boundary between runs. The pre-registration records Qwen's exploratory backfire rate as 46.8% (22 of 47); recomputing it in the pipeline that produced the results reported here gives 44.7% (21 of 47), a one-problem difference. Llama's exploratory rate is unchanged at 66.0%. The pre-registered prediction is unaffected, since both values sit far above the 33% threshold, but the same sensitivity applies in principle to the confirmatory rates: 14 of Qwen's 151 confirmatory problems have exactly zero gain. Because backfire is defined strictly as mv_gain < 0, those 14 currently count as non-backfire, so the sensitivity is one-sided and the rate can only rise. Even if all 14 crossed the boundary, Qwen's confirmatory rate would be 69.5% rather than 60.3%, still far above the 33% threshold. We report the recomputed values throughout and leave the pre-registration as tagged.

Oracle and ratio. The oracle is an upper bound, not deployable, and the fraction-of-oracle-captured ratio is noisy (small denominator), so we lead gate claims with absolute accuracy.

Entropy threshold selection. The entropy threshold was chosen by an in-sample argmax over 40 candidate values on the confirmatory 151, maximizing capture against the binary $\{N = 1, N = 64\}$ oracle relative to an $\mathrm{MV\_acc}(64)$ baseline, whereas the capture figures we report in Section 4.3 are measured against the grid oracle relative to $\mathrm{MV\_acc}(1)$. Because selection and evaluation use the same problems, the entropy gate's reported capture is optimistic.

Limited gate family (resolved in v2). v1 evaluated two verifier-free signals and noted that richer signals such as final-answer log-probability margins could behave differently, but that evaluating them requires per-token log-probability arrays which its pipeline did not store. This version stores them and evaluates that signal in Sections 4.4 and 4.5. It does not behave differently in the way that sentence hoped.

One model for the new claims. The registered second-model replication was sampled and could not be evaluated (Section 8). Cross-family replication was never purchasable either: four of five priced candidates in the 7 to 9B range refused serverless requests.

Agreement's failure has no mechanism. See Section 4.4. This is the largest unresolved question in the paper, and v2 makes it more visible rather than smaller.

The reasoning wall is a serverless result. All three models in Section 7 are downloadable, and on fixed-cost compute the caps that defeated us are affordable. That section bounds what a metered API buys at this budget, not what is knowable.

No compute-matched baseline, in v1 or v2. Every gate comparison is against a flat $N = 64$.

No margin gate was built, so the margin's failure is inferred from the signal's distribution rather than measured as a routing outcome.

An exploratory length–accuracy shape is not established. Binning problems by per-problem mean completion length gives accuracies of 0.4441, 0.3400, 0.2218, 0.3316 and 0.4098 across quintiles, but the shape-agnostic quadratic term gives $p = 0.083$. Two mechanisms were ruled out, truncation selection and within-problem bimodality. It was found on an already-exposed set with its sharpest contrast chosen after inspection, and it appears here rather than in results for that reason.

Known mechanism, narrowed in v2. Confidence failing to track correctness is an instance of documented overconfidence (Guo et al., 2017; Kadavath et al., 2022). v2 narrows what we claim from it: only one of the three signal failures is explained that way, and the unit matters (Section 4.5). Our contribution is the pre-registered, replicated, self-consistency-specific quantification and the failure of three cheap signals, not the existence of miscalibration.

7 The reasoning wall, on hosted serverless inference

v1's Limitations reported that a preliminary reasoning-native evaluation had samples exhausting the output budget on hidden reasoning, and left the question open. This section reports what happened when we tried to close it.

Scope, stated before the measurements. The claim here is about hosted serverless inference and does not hold for open weights on owned or free compute. All three models below are downloadable. Nothing here says a reasoning-native model cannot be evaluated on GPQA Diamond; it says that on a metered per-token hosted API at a small budget, three of them could not be. Serverless pricing charges per output token and reasoning-native models emit a great many, so the cost of a fixed experiment scales with the model's verbosity rather than with the size of the benchmark. The wall is a property of the billing model meeting the generation length, not of the models being hard to run. The route a follow-up should take is open weights on fixed-cost compute, where the marginal cost of an output token is zero and the binding constraint becomes wall-clock and memory: a free tier offering two 16 GB GPUs and roughly 30 GPU-hours per week is enough for a 7 to 9B model in reduced precision at long generations, though not for a 32B model without quantization or offload. Under that budget the caps that defeated us are affordable. What changes is not the science but the denominator.

Position relative to prior work. That token budgets change evaluation outcomes is established. Guedes de Souza & Panisson (2026) vary the generation budget across seven levels from 64 to 4,096 tokens, over four models and three benchmarks including GPQA Diamond at the same 198 items, 56,476 inferences in total, and report that model rankings reverse across budgets on all benchmarks ($p < 0.01$, McNemar), with 3 to 19% of items non-monotone even after controlling for truncation through a three-tier analysis. That work is stronger than this on the general claim. Two things about it define this section's scope. First, it deliberately excludes *dedicated* reasoning models, naming o1, DeepSeek-R1 and QwQ, while describing its own four as open-weight reasoning models: the exclusion is specifically of hidden chain-of-thought architectures, which is exactly our three. Second, its budget grid stops at 4,096 tokens, and one of our models returns an answer rate of 0.0000 at both 2,048 and 4,096, so across that entire grid the measurement this section reports does not exist at all.

Three models, three measured walls, each established by a real request rather than inferred from a price list. QwQ-32B is unreachable: `model_not_available`, no serverless route at any price. MiniMax-M2.7 answers, but never at a comparable cap: answer rate 0.6460 at cap 16,384 and 0.2649 at the 2,048 v1 used. Qwen3.5-9B returns nothing at any affordable cap: answer rate 0.0000 at both 2,048 and 4,096, with *mean completion equal to the cap exactly*, and 0.5938 at 8,192. The reference is Qwen2.5-7B-Instruct-Turbo at cap 2048, which answers 0.9950 over 12,672 samples. Every one of 52 samples at 2,048 and 4,096 ran to the ceiling, and at 4,096 the visible channel received 38 characters. That model's own card recommends 32,768 output tokens for general queries, so v1's 2,048 is a sixteenth of this model class's lower recommendation: this is not a model failing at a reasonable cap, it is a cap fixed before this class of model was the default.

The door, and what it changes. The provider honours a thinking control on Qwen3.5-9B. Two spellings are accepted and indistinguishable in effect: mean completion 1537.3 against 1532.8 at cap 2,048, identical truncation, no reasoning field returned either way. But a reasoning model with reasoning disabled is a second non-reasoning model. What became purchasable was not the replication that was wanted, and we do not describe it as one.

Comparability is keyed on answer rate, not on matched caps. Two models at the same cap with different length distributions produce two different output populations wearing one benchmark's name. MiniMax-M2.7 at 2,048 answers 0.2649; comparing its answered samples against Qwen's near-complete ones compares two different sets of problems. Two conditions are comparable when their answer rates match, and the answer rate is published beside every accuracy. This differs from post-hoc truncation filtering in when it applies: it is a design constraint on what to sample, not a repair applied to an existing comparison.

Cost and accounting. What this cost, how the ledger reconciles against the provider's own billing export, and how a small probe's cost projection should be calibrated are reported in Appendix A. The scientific claim of this section does not depend on them: it is that under the serverless budget available to us, these models could not be evaluated on a footing comparable to the non-reasoning models, for the three separately measured reasons above.

7.1 A gated benchmark cannot have its chains published

An infrastructure constraint we did not anticipate and found by running the check rather than by reasoning about it. Policy in this project is that benchmark question text is never committed: the sample record stores a prompt hash and never the prompt. A pre-release leakage check reduces the gated question source to hashed n-grams and scans the release tree for them. Run over the raw store it reports 5,669 hits across 198 files and blocks the release; run over the full raw release tree it reports 20,376. The cause is not a stored prompt. It is the completion text: **the model restates the question inside its own chain of thought.** The request never carried the question into storage; the response did. The consequence generalizes past this project. A chain-of-thought sample store on a gated benchmark cannot be published verbatim, however careful the storing pipeline is, because the leak is authored by the model rather than by the pipeline, and redaction rules written for what a project writes do not cover what the model writes back. This sits awkwardly beside reproducibility: the raw store is what regenerates every derived number, and it is the artifact that cannot be released. Our resolution is the two-part release recorded in the Data availability statement, a derived-only open artifact and a gated raw one on the same access terms as the benchmark. We flag it because every paper that publishes reasoning traces on a gated benchmark faces it, and we have not seen it stated.

8 When the comparability rule rejected our own registered result

The second-model replication was registered and is **not evaluated**. Two claims restated the Section 4.4 mechanism keyed on correctness rather than plurality, with thresholds 15.0 nats and 0.60 set below the exposed estimates of 21.5365 and 0.7617 so that each was a prediction, and the registration was tagged before any confirmatory sample existed. A stratification check found no dependence of either quantity on per-problem accuracy, so the thresholds were held rather than recalibrated.

Cap selection failed, and not randomly. A 32-sample probe at cap 6,144 measured answer rate 1.0000 and truncation 0.0000. The run measured 0.8931 [0.8747, 0.9090] and truncation 0.2905, with mean completion 65.3% above projection. Configuration drift was ruled out before the selection explanation was accepted: one parameter hash across all 1,253 samples matching the thinking-disabled configuration, zero nonzero reasoning-token counts, cap 6,144 throughout. The cause is measurable because the sampler iterates problem-major and so re-sampled the probe's own problems first: those 8 problems give mean completion 2082.9 and truncation 0.0312 across 128 samples, while every other problem reached gives

3546.4 and 0.3200 across 1,125. The probe reproduced itself and was precise about an unrepresentative slice, since the rest of the benchmark runs 1.703 times longer. The $k = 8$ multiplier of 22.7% could not cover a 65.3% error, because that figure describes a *random* draw of 8 problems and a fixed lowest-id slice is not one: its error is not resampleable, because there is one such slice and it is the same every time.

Consequence. At 29% truncation the pool that would be scored is selected for finishing fastest, and those samples are missing non-randomly. Scoring them would produce a number about the subset of samples that fit inside 6,144 tokens and report it as a number about the model, which is the confound the comparability rule exists to prevent. The two claims are therefore registered and unevaluated: not withdrawn, not falsified. Nor was the replication recoverable at a different cap. A right-censored fit to the 1,253 samples puts the cap required for the registered 0.9950 answer rate near 41,000 output tokens and the corresponding run at about $4.19, against $1.16 remaining; even a 0.95 rate, which would still fail the condition, requires roughly 16,268 tokens and $3.84. The replication was not lost by choosing 6,144. It was not purchasable at any cap on this budget.

The point. A methodological rule that never rejects anything is decoration. This one rejected a result its own authors had registered, sampled and paid for.

9 A reconstructed few-sample estimator

The premise was tested first, and the first formulation failed it. An estimator that infers an optimal call count from a few samples relies on the task containing a mixture of easy and hard queries: without that mixture there is nothing for it to detect. We therefore tested whether GPQA Diamond carries one for these models before testing the estimator on it. It does. **Per-problem single-sample accuracy across all 198 problems varies at 25.5 times a homogeneous null for Qwen and 17.1 times for Llama**, $p < 0.0001$ in each case over 10,000 permutations, with 45 and 37 problems respectively sitting at or below 0.05 or at or above 0.95 against a null expectation near zero. GPQA Diamond is hard on average and is *not* uniformly hard for these models: it carries exactly the easy/hard mixture Chen et al. (2024) rely on. The estimator is therefore tested where the mixture is present and measured, rather than where its premise is absent, and a negative result here is informative about the estimator rather than about the benchmark.

Mixture presence turned out to be necessary but not sufficient. The mixture is present, and the non-monotonicity it is supposed to produce is not: the confirmatory aggregate accuracy curve across the grid spans 0.52 points and does not fall. Whatever makes an aggregate curve rise and then decline, the presence of a per-problem difficulty mixture does not guarantee it, at least at this benchmark's difficulty and this scale of model.

We reconstructed the few-sample call-count estimator that follows from Chen et al. (2024) and pre-registered a comparison against a naive within-k baseline on v1's confirmatory 151, using paired per-problem regret where negative favours the estimator. At $k = 8$ the difference is -0.0213 $[-0.0335, -0.0091]$, a PASS by more than the registered resolution floor of 0.0161. At $k = 4$ it is -0.0186 $[-0.0332, -0.0041]$, a PASS that was registered as underpowered. At $k = 16$ it is -0.0040 $[-0.0085, +0.0004]$, a **FAIL** by 0.0004, reported as a failure rather than rounded into the pattern of the other two. A shrinkage baseline was added post hoc and is labelled as such: at $k = 4$ and $k = 16$ every shrinkage strength is worse than the plain baseline, and at $k = 8$ the weakest helps slightly. It is not registered and decides nothing.

10 Corrections and disclosures

Recorded as a section rather than a footnote.

1. **A superseded draft was cited** in place of the published preprint in project documentation. A test now scans every document in the repository for citations to superseded sources.

2. **Three per-problem properties were described as independent** before being tested. Within one model at $n = 127$, per-problem accuracy against the sub-2-nat margin tail gives -0.2614 $[-0.3912, -0.1156]$, excluding zero.

3. **The margin run changed concurrency mid-run.** Because the sampler iterates problem-major, the resulting regime comparison is between problems rather than within them, so it is confounded and decides nothing. Three manifests are marked as reconstructed rather than contemporaneous.

4. **An option-order claim was corrected after tagging.** Options are shuffled by a row-index-keyed generator, reproducing v1's shuffle; prompt hashes are equal across both model stores for all 198 problems, verified.

5. **An answer-rate pairing rule did not cover prose tables** until a quintile table of accuracies was published without them.

6. **The second-model cap was chosen from 8 non-random problems** (Section 8). Probe problems are now drawn at random with a recorded seed.

7. **v1's own split has a leak:** 3 of its 50 exploratory ids sit inside the confirmatory 151. Disclosed here; v1's verdicts are not recomputed.

8. **A bibliography entry's authors were written from memory.** The entry for a length-invariant uncertainty estimator was added from a search-result title rather than from the paper: its author list named a person who is not an author and omitted two who are. It survived a merge, a build and a full pre-submission read, and was caught in an entry-by-entry audit asking which references had actually been opened. Corrected against the arXiv record. **The root cause is the one this paper has disclosed twice already**: a claim made from recollection rather than from a source, of the same kind as citing a superseded draft and as asserting three properties were independent before testing them. It is now a check rather than a habit: every entry carrying an arXiv id is compared against the arXiv API for title, full author list and year, and every entry without one must record which source was opened and when, with an entry lacking that record failing the suite. All sixteen entries, including the eleven inherited from v1, were re-verified this way; no further mismatch was found.

9. **v1's Setup describes uniform-random tie-breaking; its code breaks ties lexicographically.** Found while reproducing v1's figures from the archived derived data: the plurality function sorts the tied answers and returns the first. The published numbers follow the code. Quantified rather than only stated: Qwen has 5 full-pool plurality ties of 198 and Llama 2, and counting $N = 64$ subsets Qwen has 47 where the correct answer is in a tie while **Llama has none**, so Llama's figures are convention-independent outright. Lexicographic order is not a biased choice on this dataset: the correct answer is A on 48 problems, B on 51, C on 47 and D on 52 after the shuffle, $\chi^2 = 0.343$ on 3 degrees of freedom. Across 5,000 uniform-random tie-break seeds Qwen's MV_acc(64) has mean 0.3726 and standard deviation 0.0051, spanning 0.3619 to 0.3833, and the lexicographic 0.3686 sits at the 31.7th percentile: inside the distribution, slightly unlucky, not an outlier. Backfire spans 109 to 113 of 198. **The convention is a third-decimal effect and changes no conclusion:** every seed leaves backfire far above the 33% pre-registered threshold and still a majority, and every seed leaves MV_acc(64) above MV_acc(1) of 0.3419. v1's verdicts are not recomputed.

10. **v1's Positioning describes the benchmark as uniformly hard; the stored data does not support that for these models.** Per-problem single-sample accuracy varies at 25.5 times a homogeneous null for Qwen and 17.1 times for Llama, $p < 0.0001$ over 10,000 permutations, with 45 and 37 problems at or below 0.05 or at or above 0.95. GPQA Diamond is hard *on average* and is not uniform in difficulty for these models. **This touches no v1 result and no v1 number**: it bears only on the positioning against Chen et al. (2024), whose account turns on exactly that mixture. It is disclosed rather than quietly corrected because Section 9 builds on the same positioning, and using a premise in one section while leaving its overstatement

standing in another is the kind of quiet reliance this section exists to prevent. The sentence in Section 5 is qualified accordingly.

11 Conclusion

On hard reasoning problems, self-consistency backfires on the majority of problems, pre-registered and confirmed on the full GPQA Diamond benchmark for Qwen and corroborated by a second family from a near-chance baseline. **That result is unchanged from v1.**

What has changed is the explanation for why cheap verifier-free gates cannot recover the headroom. v1 attributed it to confidence not tracking correctness. With a third signal measured at the right place, that account no longer covers the evidence. The entropy gate was reading a diluted statistic rather than a confidence signal. The answer-token margin, read where dilution cannot apply, fails because the answer token is saturated: a sample that contradicts its own problem's plurality is as committed as one that agrees with it, so the variance a router would need has already been spent upstream. And the agreement gate's failure is explained by neither, and remains open. Recovering the headroom likely requires a signal external to the model's own samples, or one read earlier in the chain than the answer. Whether reasoning-native models escape any of this remains open, and on hosted serverless inference at a small budget it is not measurable at all.

A Cost, accounting, and probe calibration

Lifted out of Section 7 so that the body carries the scientific claim and this carries the bookkeeping. Nothing is omitted.

The bill, reconciled against the provider. Our client-side ledger records \$4.697153 across 14,125 sampled requests. The provider's own billing export, filtered to this project's API key and summed from unrounded quantities times unit prices rather than the rounded per-line amounts, gives **\$4.701169** across six line items over two days. The difference is **\$0.004017, or 0.085%**, and the ledger is the lower of the two. Guedes de Souza & Panisson (2026) report 56,476 API calls and no monetary cost. We report both figures because the negative results in this paper are only interpretable alongside what was affordable, and because a cost claim that is not reconciled against the counterparty is an assertion rather than an audit.

Where the 0.085% goes, and why it is one-sided. The residual is about 4,600 unrecorded tokens on one model and 11,000 on the other. A client-side ledger writes a row when a response *arrives*, so it cannot see a request that was issued, billed, and then abandoned. This run has three such boundaries: a mid-run concurrency change, a `SIGTERM` that stopped the second-model run at concurrency 16, and an interrupted probe. Each leaves its in-flight requests billed by the provider and absent from the ledger. **This is a property of the accounting method rather than a discrepancy to explain away**: a client-side ledger undercounts any interrupted run by exactly its in-flight requests, always in the same direction, and the size of the undercount is the concurrency times the number of interruptions. Anyone reporting costs this way should expect a small one-sided gap and should say so rather than round it off.

The reconciliation is only possible because the key was dedicated. Project security policy required a separate API key per project, for revocation rather than for accounting. The same billing account carries a second key with \$0.292479 of unrelated spend on 10 and 11 August, including a model this project never used, and all twelve line items across both keys sum to \$4.99. With a shared key the Argmax bill would not have been separable from that at all: no filter, no reconciliation, and the \$4.99 would have had to be reported as the project's cost or not reported. **A security rule turned out to be an accounting rule**, and we would not have discovered that had the second key's spend not happened to overlap in time.

Projecting cost from a small probe. Mean completion length is a per-problem property at 119.66 times the variance a homogeneous null produces, so a probe over k problems inherits that between-problem spread and taking more samples per problem does not reduce it. Resampling the 198 per-problem means, 20,000 trials per k, the multiplier needed for a projection to cover the truth 95% of the time is 34.9% at $k = 4$, 22.7% at $k = 8$, 15.2% at $k = 16$ and 10.0% at $k = 32$. Probes are near-unbiased in the median, yet just over half underestimate at every k because per-problem means are right-skewed, so a ceiling set at the projection is wrong about half the time.

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